

Filing Lens

Cited answers from company filings, from a local 3B model (Qwen 2.5, runs on the phone)

Retail investors are handed 100-page 10-Ks. Chatbots answer anyway, with no page reference, and the filing gets uploaded to someone else's server.

FinanceBench (Islam et al., 2023, arXiv:2311.11944): GPT-4-Turbo with a retrieval system **incorrectly answered or refused 81%** of 150 open-book filing questions.

Problem: wrong answers look the same as right ones, and privacy is the price of asking.

Demo

lower earnings and higher incentive compensation payments in the current year as a result of strong fiscal 2022 results, partially offset by p.29 cash equivalents were as follows (\$ in millions): January 28, 2023 January 29, 2022 Cash and cash equivalents \$ 1,874 \$ 2,936 The decrease in cash and cash equivalents in fiscal 2023 was primarily driven by share repurchases, capital expenditures and dividend payments. p.29

93%

WELL SUPPORTED

0

ABSTAIN BELOW 50

100

Calibrated probability that the retrieved pages support an answer.

Extractive preview (no model downloaded) · no verifier · 1363 ms · citation coverage 100% · sources: p.29

Ask about this filing...

ASK

lower earnings and higher incentive compensation payments in the current year as a result of strong fiscal 2022 results, partially offset by p.29 cash equivalents were as follows (\$ in millions): January 28,

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CLOSE

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PAGE

TEXT

>

EVIDENCE HIGHLIGHTED

Liquidity and Capital Resources

The decrease in cash and cash equivalents in fiscal 2023 was primarily driven by share repurchases, capital expenditures and dividend payments. These decreases were partially offset by cash flows from operations, primarily from its services.

Cash and cash equivalents were as follows (\$ in millions)

	January 28, 2023	January 29, 2022
Cash and cash equivalents	\$ 1,874	\$ 2,936

Cash flow provided by (consumed)

	2022	2021	2020
Operating activities	\$ 654	\$ 526	\$ 439
Investing activities	\$ 661	\$ (375)	\$ 696
Financing activities	\$ 661	\$ (375)	\$ 696
Effect of exchange rate changes on cash	\$ (1)	\$ (1)	\$ (1)
Change in cash and cash equivalents	\$ 654	\$ 145	\$ 123

Operating Activities

The decrease in cash provided by operating activities in fiscal 2023 was primarily driven by the timing and amount of compensation payments in the current year as a result of strong 2022 results, partially offset by the timing and amount of inventory purchases and payments.

Investing Activities

The decrease in cash used in investing activities in fiscal 2023 was primarily driven by the purchases of Current Health and Wellness in fiscal 2023 and the decrease in purchases of investments, partially offset by an increase in capital spending for initiatives to support our business.

Financing Activities

The decrease in cash used in financing activities in fiscal 2023 was primarily driven by lower share repurchases.

Source of Liquidity

From time to time, we may need to raise capital to fund our operations, capital expenditures, share repurchases, dividend and strategic initiatives. We have a variety of financing options available to us, including bank credit facilities, commercial paper, corporate debt, and equity offerings. We have a strong relationship with our lenders and we believe we have the ability to obtain additional financing, if necessary, on favorable terms.

We have a \$1.25 billion revolving credit facility that expires in May 2025. There were no borrowings outstanding under the Five Year Facility Agreement as of January 28, 2023, or January 29, 2022.

Our ability to continue to access the Five Year Facility Agreement is subject to our compliance with its terms and covenants, including financial covenants. The financial covenants require us to maintain certain financial ratios. Our financial ratios for 2022 are in compliance with all financial covenants. If a portion of a debt were to issue with respect to any of our other debt, it would be subject to the Five Year Facility Agreement as well.

support an answer.

Extractive preview (no model downloaded) · no verifier · 1363 ms · citation coverage 100% · sources: p.29

02

QUESTION

What was Walmart's capital expenditure in fiscal 2023?

NOT IN THIS FILING

The retrieved pages don't look like they answer this, so the model wasn't asked.

30%

NOT SUPPORTED

0

ABSTAIN BELOW 50

100

Calibrated probability that the retrieved pages support an answer.

CLOSEST PAGES, IF YOU WANT TO CHECK YOURSELF

p.30

p.29

p.61

Stopped at the feature gate · 32 ms on-device

Ask about this filing...

ASK

Cited answer + confidence ruler Page tab opens the PDF, cited sentences highlighted Wrong company → NOT IN THIS FILING

Screens from the case-file UI at main `dd0071e`: headless Chromium, 390 px wide, extractive preview (no model downloaded, so the app quotes the filing). Launch video: `brag-output/brag.mp4`. Phone recording: **[ASK: phone recording link]**.

Privacy proof

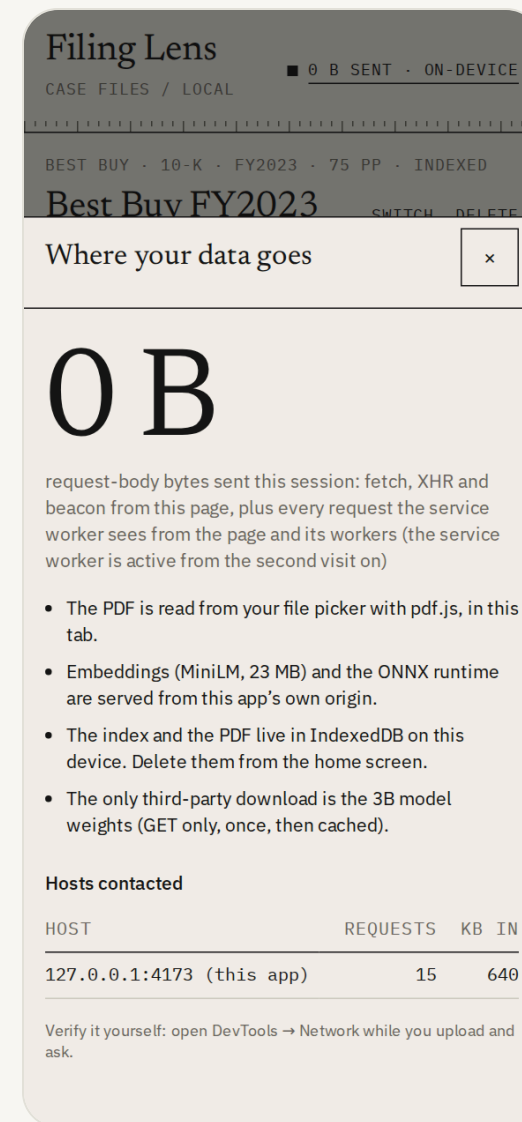
The document never leaves the device. The build's e2e test enforces this; it isn't just claimed.

Requests during upload → index → ask → abstain **24**

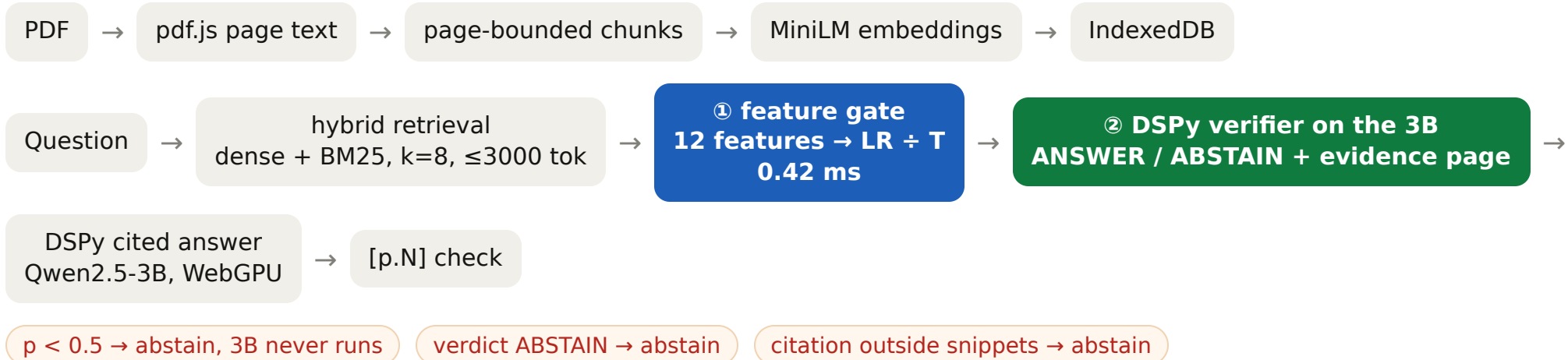
to any host other than the app's own origin **0**

carrying a request body (POST/PUT/beacon) **0**

- pdf.js, MiniLM embeddings, ONNX runtime, wllama: npm packages served from the app origin, no external scripts
- Index + PDF stored in IndexedDB
- Only third-party fetch: the Qwen2.5 weights (GET, once, then cached). Airplane mode after that: **[ASK: confirm on phone]**



Architecture: a two-stage cascade



① is cheap and catches wrong-company questions (it scores retrieval and word coverage, not content). ② reads the actual sentences: it is a DSPy program, compiled with BootstrapFewShot(2 demos); MIPROv2 light tried, dev 0.519 < 0.548, not kept on Qwen/Qwen2.5-3B-Instruct fp16 via llama.cpp (Kaggle T4; vLLM failed to start), exported to `app/prompts/verifier.json` and rebuilt in the browser byte-for-byte. A unit test checks this against DSPy's own rendering.

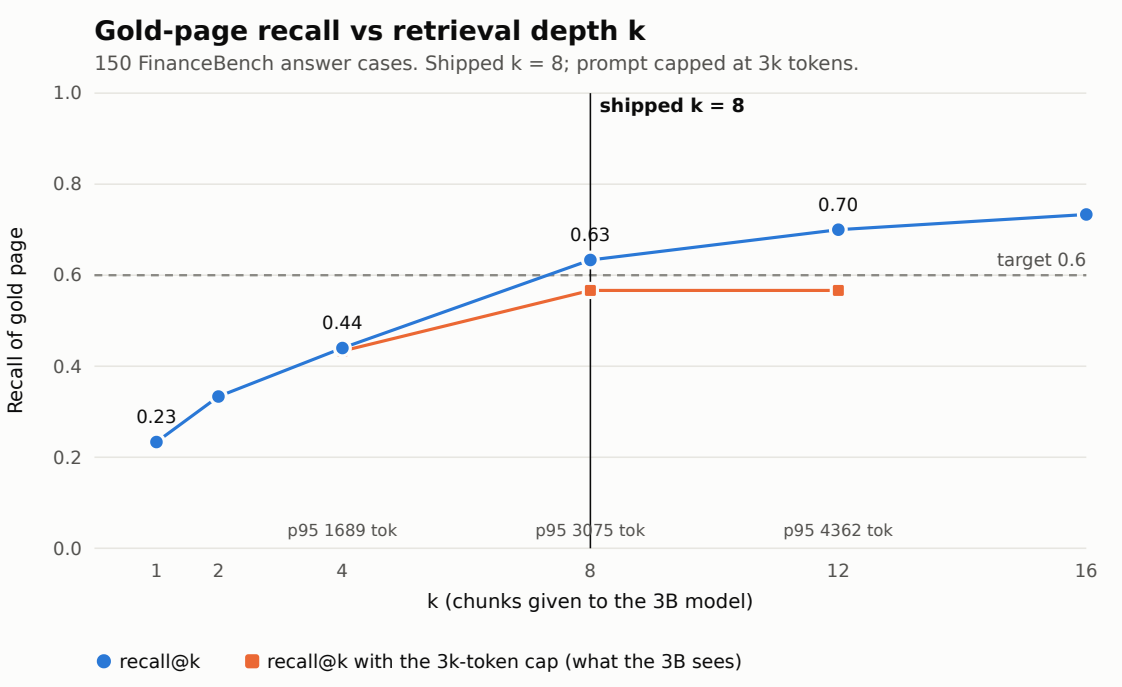
Bench: gate vs verifier vs cascade

FinanceBench open-source (patronus-ai/financebench), 150 questions, 84 filings. Gold answer = question on its own filing; **Easy** abstain = another company's 10-K; **Hard** abstain = own filing with the gold pages (and any page still holding the gold numbers) removed; 10 questions excluded where the answer was still retrieved. Test split (company-grouped), 160 instances.

System	Easy acc.	Hard acc.	ECE E / H	Cite page	p50	3B calls
Feature gate only	0.864	0.524	0.192 / 0.156	0.279*	0.42 ms	0%
Verifier only (fp16_llamacpp)	0.509	0.495	0.476 / 0.496	0.500	7447 ms	100%
Cascade: gate → verifier (fp16_llamacpp)	0.518	0.495	0.392 / 0.404	0.500	7202 ms	53%
Verifier only (gguf_q4)	0.527	0.495	0.468 / 0.474	0.500	7156 ms	100%
Cascade: gate → verifier (gguf_q4)	0.536	0.495	0.410 / 0.391	0.500	6939 ms	53%

* gate proposes no page: top-ranked retrieved page. Gate p50 measured in Node on Node v22.22.0, 4 vCPU container, no GPU; verifier p50: one verifier call on a Kaggle T4 (llama.cpp; vLLM failed to start).

Gate, all 440 out-of-fold: Easy acc 0.870, abstain P/R 0.828 / 0.933; Hard acc 0.524. Two "does the top passage answer this?" features: Hard 0.528 → 0.524.



Limits (measured)

- **The 3B verifier abstains too much; off by default** (opt-in Strict mode).
- **The feature gate can't see content: Hard 0.524** (always answering scores 0.517 on this set). It scores retrieval and word overlap, so "the gold page" and "a nearby page about the same metric" look alike. The verifier stage was meant for that case. It didn't help: the compiled 3B verifier abstains on almost everything (Hard 0.495, Easy 0.527 on Q4; auto-graded answer accuracy 0.036). Behind the gate it cuts Easy accuracy from 0.864 to 0.536, so it ships off by default, as an opt-in Strict mode.
- **Retrieval is the bottleneck:** gold-page recall@4 0.440, @8 0.633, @12 0.700. With the 3k-token cap the 3B sees the gold page 57% of the time; the sentence snippets keep it 53%.
- **A failure I shipped and fixed:** Best Buy's 10-K never mentions Walmart, yet "Walmart capex?" passed the first gate. Adding a feature for whether the companies a question names appear in the filing at all fixed it (gate accuracy 0.82 → 0.91 on the earlier own-vs-wrong-company bench).
- Answers are auto-graded (number within 1%, else token-F1), an approximation of FinanceBench's human grading. Scanned PDFs (no text layer) are rejected.

Roadmap

- Verifier: 2 bootstrapped demos on a 3B model learned to abstain. Next: more labelled demos, a threshold on $P(\text{ANSWER})$ tuned on dev rather than the argmax verdict, and a 7B check on a laptop GPU
- Retrieval: table-aware chunking for financial statements; a reranker only if one ships as an npm package (none found) and adds <150 ms
- Raise the context budget if the phone allows (uncapped k=8 prompts reach 3586 tokens)
- Indian filings: BSE/NSE annual reports, Hindi questions

The ask

A Finale slot, one iQOO test phone to benchmark on-device inference, and mentor feedback on the Snapdragon NPU path.

Repo: github.com/rajyyug1132/Filing-Lens-Private-Cited-Answers

Cited answers, or an honest "I can't tell". Your filing stays yours.